

Spectral energy distribution of a starburst galaxy

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In this assignment, you are going to assemble the spectral energy distribution (SED) of a starburst ultra-luminous infrared galaxy (ULIRG) from several different radiative processes.

1. The data table for the galaxy M82 will be provided. You need to read in the frequency (Hz) and flux density S_ν in arbitrary units (units comparable to $[J s^{-1} m^{-2} Hz^{-1}]$).
 - (a) Plot M82 flux density in the given units as a function of wavelength or frequency.
2. The spectral energy distribution over this wavelength range has contributions from four main processes: starlight, **dust emission**, **thermal free-free emission** and **synchrotron**. You are going to write functions to generate spectra for these three processes. Then you need to match the observed M82 spectrum (e.g. with a χ^2 statistical fitting) to get the associated parameters. Have a look at [Fig 5 \[1\]](#), (*Downes et. al. 2007*) to get an idea.
3. In thermal equilibrium, dust emits like a “modified black-body”.

$$S_\nu = \frac{M_{\text{dust}}}{D^2} \kappa_\nu B_\nu(T_{\text{dust}}), \quad (1)$$

where S_ν is the flux density, M_{dust} is the dust mass, $B_\nu(T_{\text{dust}})$ is the Planck function, D is the distance to the galaxy M82, and κ_ν is the absorption opacity. κ_ν can be derived from the composition of grains and Mie scattering cross section Q_{abs} as follows:

$$\kappa_\nu = \frac{3Q_{\text{abs}}}{4a\rho}, \quad (2)$$

where a is the grain size and ρ is the grain mass density. You can find the optical properties of various dust compositions at this link - <https://www.astro.princeton.edu/~draine/dust/dust.diel.html>. We will explore the contribution of two sizes of dust grains, which contributes to the spectrum at different wavelength ranges.

- We consider large dust grains like Silicates with radii between $0.001 - 10 \mu\text{m}$.
- The $3 - 30 \mu\text{m}$ region of the spectrum has a series of wiggles which are the features from polycyclic aromatic hydrocarbon (PAH) vibrational bands dominating the spectrum. The PAHs are smaller graphite particles with radii between $0.000355 - 0.0010 \mu\text{m}$.

For both cases, the spectrum follows equation (1) with different dust temperature, mass, and opacity.

- (a) Plot the absorption opacity for PAHs as a function of frequency.
- (b) Plot the M82 data points and the scaled dust spectrum for both cases after fitting it with the observed spectrum.
- (c) Over what wavelength range does dust emission dominate M82's spectrum?
- (d) What can you say about the temperature of Silicates and PAHs just by looking at the M82 spectrum? (*Hint: In your argument, combine what you know about Mie Scattering and Wien's Law.*)
- (e) By choosing some grain composition and size, what temperature does your model need to match the observed dust emission from M82 for both cases? Compare this temperature with what is found in standard literature.

(f) What is the region of electromagnetic spectrum where dust/PAHs dominate?

4. The thermal Bremsstrahlung or free-free spectrum for M82 follows a power law given by

$$P_{\text{ff}} \propto (T_e)^{-1/2} \exp(-h\nu/kT_e) \bar{g}_{\text{ff}}(T_e, \nu), \quad (3)$$

where T_e is the electron temperature in HII regions and $\bar{g}_{\text{ff}}$ is the Gaunt factor, often quoted to be unity. (Refer to *Rybicki & Lightman* or any other references for the full spectra). However, at low frequency, $\bar{g}_{\text{ff}} \neq 1$ and the spectrum becomes

$$P_{\text{ff}} \propto (T_e)^{-1.35} \nu^{-2.1} B_{\nu}^{\text{RJ}}(T) \quad (4)$$

where $B_{\nu}^{\text{RJ}}(T)$ is the Rayleigh-Jeans Planck function.

- (a) Choose T_e appropriately [2](*Schober et. al. 2016*) and scale assuming fixed values of the quantities in equation (5.14a) from the book.
 - (b) Plot the M82 data points and the scaled free-free spectrum after fitting.
 - (c) Over what region of electromagnetic spectrum does free-free make an important contribution to M82's spectrum?
5. Synchrotron emission is the radiation emitted by relativistic particles accelerated by a magnetic field of strength $|\vec{B}|$. The synchrotron spectrum has a characteristic is given as a power-law distribution

$$P_s(\nu) \propto C B^{(p+1)/2} \nu^{-(p-1)/2} \propto C B^{a+1} \nu^{-a}. \quad (5)$$

p and C are the power-law index and ratio for distribution of particles of energy E as in *Rybicki & Lightman* [3]. $a = (p - 1)/2$ is the synchrotron spectral index. Refer to [3] (*Rybicki & Lightman*) for the full form of the equation.

- (a) You need to fit the observed spectrum assuming C , B and pitch angle α (angle between the the velocity field $\vec{v}$ and $\vec{B}$) to match it to the distribution of certain frequency range of M82's spectrum.
 - (b) What power-law slope a do you need to reproduce M82's synchrotron spectrum?
 - (c) Plot the M82 data points and the scaled synchrotron spectrum. Quote the scaling factor used.
 - (d) Find the slope of the synchrotron spectrum directly from the M82 spectrum. *Hint: Note down the luminosities at two different frequencies for the synchrotron spectrum and use the formula for slope of a straight line.* Compare it with the answer obtained in part (b).
 - (e) Over what region of electromagnetic spectrum does synchrotron dominate?
6. Final plot and results.
- (a) Make a script that generates one final plot with the spectra for all three radiative processes on top of the M82 data points. Match it with the template provided.
 - (b) Share you python code, all figures and report (*in .pdf format*) answering the questions above. I would expect your code to work on any system running Python and produce the same final plot as in your report.
 - (c) Please work on your code and report individually; similarities between any two reports or the main codes will be penalised.

References

- [1] Downes, D. and Eckart, A. Black hole in the west nucleus of arp 220. *A&A*, 468(3):L57–L61, 2007.
- [2] Jennifer Schober, D. R. G. Schleicher, and R. S. Klessen. Tracing star formation with non-thermal radio emission. *MNRAS*, 468(1):946 – 958, June 2017.
- [3] George B. Rybicki and Alan P. Lightman. *Radiative processes in astrophysics*. 1979.